

GIOVANNI QUINONES-VALDEZ

giovas@ucla.edu | (+1) 352 256 4917 | <https://giovanniquinones.github.io>

RESEACH INTEREST

My research seeks to determine how genetic and post-transcriptional variants regulate the transcriptome, including gene expression, splicing, and polyadenylation, through the development of computational and statistical methods for multi-modal sequencing data. By enabling these analyses at scale, my work aims to decode the regulatory logic of RNA processing and its disruption in neurological disease, particularly Alzheimer's Disease.

EDUCATION

- **University of California, Los Angeles** | Bioengineering (Bioinformatics focus) | Ph.D. 12/2021
Advisor: Dr. Xinshu (Grace) Xiao
Dissertation: "Function and Regulation of Nucleotide Variants in RNA"
- **University of Florida** | Biological Engineering | B.S. 05/2015
Minor: Biomolecular Engineering

EXPERIENCE

- **Postdoctoral Fellow** | *Department of Integrative Biology & Physiology, UCLA* 01/2022 – Present
My research integrates statistical method development (isoLASER, scAllele) with multi-modal genomics data analysis (spanning long-read, single-cell, and bulk RNA sequencing) to study the role of RNA variants in basic transcriptomic regulation and disease.
- **Collaboratory Fellow** | *Institute for Quantitative & Computational Biosciences, The Collaboratory, UCLA* 01/2022 – Present
I provided computational research support to UCLA laboratories through consulting and data analysis, and trained postdoctoral scientists and graduate students in bioinformatics through hands-on workshops.
- **Part-time Scientific Consultant** | *Rattan Life Science* 08/2025 – 09/2025
Delivered hands-on training to staff scientists in the processing and analysis of single-cell RNA-sequencing (scRNA-seq) data analysis using R and artificial intelligence (AI) approaches for genomics data analyses.

PUBLICATIONS

1. King AJ, Amoah K, Zhang L, Bahn JH, Barney RM, **Quinones-Valdez G**, Xiao X. *Allele-specific splicing in the human brain reveals genetic variants linked to protein isoforms and Alzheimer's disease.* *Genome Research.* <https://doi.org/10.1101/gr.281117.125>. (2026)
2. **Quinones-Valdez G**, Chang JW, Magaki SD, Vinters HV, Salamon N, Liao L, Wang AC, Fallah A, Owens GC. *Immune landscape of the affected brain in Rasmussen encephalitis.* *Scientific Reports.* <https://doi.org/10.1038/s41598-026-51295-3>. (2026).

3. **Quinones-Valdez G**, Amoah K, Xiao X. *Long-read RNA-seq demarcates cis- and trans-directed alternative RNA splicing*. *Nature Communications*. <https://doi.org/10.1038/s41467-025-64605-6>. (2025)
4. Hervoso JL, Amoah K, Dodson J, Choudhury M, Bhattacharya A, **Quinones-Valdez G**, Pasaniuc B, Xiao X. *Splicing-specific transcriptome-wide association uncovers genetic mechanisms for schizophrenia*. *The American Journal of Human Genetics*. <https://doi.org/10.1016/j.ajhg.2024.06.001>. (2024)
5. Zheng R, Dunlap M, Lyu J, Gonzalez-Figueroa C, Bobkov G, Harvey SE, Chan TC, **Quinones-Valdez G**, Choudhury M, Vuong A, Flynn RA, Chang HY, Xiao X, Cheng C. *LINE-associated cryptic splicing induces dsRNA-mediated interferon response and tumor immunity*. *Molecular Cell*. <https://doi.org/10.1016/j.molcel.2024.05.004>. (2024)
6. Liu Z, **Quinones-Valdez G**, Fu T, Choudhury M, Reese F, Mortazavi A, Xiao X. *L-GIREMI uncovers RNA editing sites in long-read RNA-seq*. *Genome Biology*. <https://doi.org/10.1186/s13059-023-03012-w>. (2023)
7. **Quinones-Valdez G**, Fu T, Chan WT, Xiao X. *scAllele: a versatile tool for the detection and analysis of variants in scRNA-seq*. *Science Advances*. <https://doi.org/10.1126/sciadv.abn6398>. (2022)
8. Chan TC, Fu T, Bahn JH, Jun HI, Lee JH, **Quinones-Valdez G**, Cheng C, Xiao X. *RNA editing in cancer impacts mRNA abundance in immune response pathways*. *Genome Biology*. <https://doi.org/10.1186/s13059-020-02171-4>. (2020)
9. **ENCODE Consortium**. *Perspectives on ENCODE*. *Nature*. <https://doi.org/10.1038/s41586-020-2449-8>. (2020)
10. **ENCODE Consortium**. *ENCODE Phase III: Building an Encyclopedia of Regulatory Elements for Human and Mouse*. *Nature*. <https://doi.org/10.1038/s41586-020-2493-4>. (2020)
11. **Quinones-Valdez G**, Tran S, Jun HI, Bahn JH, Yang EW, Zhan L, Brummer A, Wei X, VanNostrand E, Pratt G, Yeo GW, Graveley BR, Xiao X. *Regulation of RNA editing by RNA Binding Proteins in Human Cells*. *Communications Biology*. <https://doi.org/10.1038/s42003-018-0271-8>. (2019)
12. Yang EW, Bahn JH, Hsiao YH, Tan BX, Sun Y, Fu T, Zhou B, Van Nostrand EL, Pratt, Freese P, Wei X, **Quinones-Valdez G**, Urban AE, Graveley BR, Burge CB, Yeo GW, Xiao X. *Allele-specific binding of RNA Binding Proteins Reveal Functional Genetic Variants in the RNA*. *Nature Communications*. <https://doi.org/10.1038/s41467-019-09292-w>. (2019)
13. Hsiao YH, Bahn JH, Yang Y, Lin X, Tran S, Yang EW, **Quinones-Valdez G**, Xiao X. *RNA editing in nascent RNA affects pre-mRNA splicing*. *Genome Research*. <https://doi.org/10.1101/gr.231209.117>. (2018)

ARTICLES UNDER REVIEW

1. Zheng R, **Quinones-Valdez G†**, Xiao X, Cheng C. *CryEx.v2: An enhanced protocol for the quantitative detection and visualization of cryptic splicing using transcriptome data*. *STAR Protocols*
2. Barney RM, **Quinones-Valdez G**, King AJ, Amoah K, Wang W, Xiao X. *Allele-specific alternative polyadenylation links noncoding genetic variation to Alzheimer's disease risk*. *Genome Biology*.
3. Vavilina A, Calvanese V, Gu Y, Ma F, **Quinones-Valdez G**, Perrod C, Colombo G, Fares I, Ko MW, Kaaja I, Julia, Aguade-Gorgorio J, Nadel B, Liebscher S, Wang Y, Goodridge H, Crooks GM, Pellegrini M, Xiao X, Schenke-Layland K, Mikkola HKA. *Developmentally regulated long and short isoforms of MLLT3 balance human hematopoietic stem cell fate decisions*. *Cell Stem Cell*.
4. **ENCODE Consortium**. *The Encyclopedia of DNA Elements*. *Nature*

CONFERENCE ABSTRACTS

1. Kaaja I, Wang Y, Kilpivaara O, Wartiovaara-Kautto U, **Quinones-Valdez G**, Gu Y, Turmon A, Ma F, Wang Y, Aguade-Gorgorio J, Ko MW, Grimbert S, Wang Y, Katainen L, Park S, Pellegrini M, Xiao X, Seet CS, Chang VY, Arboleda V, Backus K, Mikkola HKA. *Tracking the Stemness Mechanisms in Refractory Mecom Rearranged Acute Myeloid Leukemias at Single Cell Resolution*. European Hematology Association. 2025.
2. Frenz-Wiessner S, Wang Y, Rohde Y, **Quinones-Valdez G**, Issa H, Gonçalves-Dias J, Schuschel K, Mack P, Ko MW, Wang Y, Ma F, Sakamoto K, Chang V, Xiao X, Klusmann JH, Mikkola HKA. *Single-cell tracing of myeloid leukemia evolution in down syndrome reveals a latent progenitor bridging disease onset, progression and relapse*. Blood. 2025
3. Biswal MR, Ildenfonso CJ, Rossmiller BP, Mao H, Li H, Zhu P, Tong Y. **Quinones-Valdez G**, Lewin AS. *Delaying Retinal Degeneration in a Mouse Model of Geographic Atrophy: An Antioxidant Gene Therapy Approach*. Investigative Ophthalmology and Visual Science. 2015.
4. Biswal MR, Ildenfonso CJ, Wang Z, Mao H, Li H, Zhu P, **Quinones-Valdez G**, Lewin AS. *Effectiveness of Antioxidant Gene Therapy in Delaying Retinal Degeneration*. Molecular Therapy. 2015.

TEACHING AND MENTORING

- **Instructor for the workshop “Unix Command Line”** 01/2026 – 04/2026
Institute for Quantitative and Computational Biosciences, UCLA (Quarterly)
- **Instructor for the workshop “Single Cell RNA-Seq. Data Analysis with R”** 01/2022 – 03/2025
Institute for Quantitative and Computational Biosciences, UCLA (Quarterly)
- **Instructor for the Bruins in Genomics (BIG) summer program** 07/2024
Institute for Quantitative and Computational Biosciences, UCLA 07/2023
07/2020
- **Guest lecturer for the course “Intro to High-Throughput Data Analysis”** 03/2024 – 06/2024
Department of Biomathematics, UCLA 03/2023 – 06/2023
- **Assistant instructor for the course “Concepts in Molecular Biosciences”** 01/2024 – 03/2024
Department of Molecular Biology, UCLA
- **Teaching Assistant for the course “Molecular Parasitology”** 09/2017 – 12/2017
Department of Microbiology, Immunology, and Molecular Genetics, UCLA
- **Member of the Xiao Lab at UCLA** 01/2016 – Present
 - Mentored junior graduate and undergraduate students.
 - Interviewed and hired students to aid with computational analysis.
 - Led 2 undergraduate students to first authorship on papers in Genome Biology and Genome Research

AWARDS

- **ASHG Trainee Research Excellence Award winner** 11/2024
Awarded by the American Society of Human Genetics (ASHG) for outstanding

trainee research presentation and scientific contribution at the annual meeting.

- **QCBio Collaboratory Postdoctoral Fellowship** 02/2022 – 03/2025
Prestigious funding mechanism available to postdocs and project scientists conducting research in computational biology and bioinformatics.
- **Young Bolivian Abroad Award** 12/2018
Recipient of the tarijano abroad award (diaspora category), awarded by the regional government of Tarija, in recognition of academic achievement and contributions as a representative of Tarija abroad.
- **UCLA Bioengineering Supplementary Fellowship** 03/2017
Fellowship support awarded by the UCLA department of bioengineering to support graduate research and training.
- **UCLA Bioengineering Research Fellowship** 01/2016
Competitive graduate fellowship supporting research in bioengineering.
- **Davis Scholarship for Undergraduate Education** 08/2011 – 05/2015
Merit-based scholarship awarded through the Davis United World College Scholars Program, supporting graduates of the United World Colleges network.

RESEARCH TALKS

- **Title: “Long-read RNA-seq demarcates cis- and trans-directed alternative RNA splicing”**
 - **American Society of Human Genetics, *Featured Plenary Talk*** 11/2024
 - **UCLA Riboforum, *Selected Talk*** 01/2024
 - **International Society for Computational Biology, *Selected Talk*** 11/2023
 - **UCLA QCBio Retreat, *Selected Talk*** 09/2023
- **Title: “scAllele, a versatile tool for the detection and analysis of variants scRNA-seq”**
 - **Open Box Science, *Invited Talk*** 04/2023
 - **ENCODE Consortium Retreat, *Talk*** 03/2022

JOURNAL REVIEWER

- **Genome Biology** 11/2025 – 10/2026
- **BMC Genomics** 01/2026
- **Genomics, Proteomics, and Bioinformatics** 08/2024
- **Communications Biology** 03/2023

SOFTWARE DEVELOPMENT

- **isoLASER – Lead Developer** 10/2025
Built in Python | GitHub: <https://github.com/gxiaolab/isoLASER>, PyPi

isoLASER performs variant calling, haplotyping, and allele-specific splicing analysis in long-read RNA-Sequencing data. This one-stop solution enables the classification of genetically regulated splicing events revealing transcriptome-wide patterns of regulation and disease-relevant isoform regulation such as the HLA gene family and Alzheimer's disease risk genes.

- scAllele – Lead Developer** 09/2022
Built in Python | GitHub: <https://github.com/gxiaolab/scAllele>, PyPi
 scAllele is an integrative tool that performs high-confidence calls of small variants in single-cell RNA-Sequencing using a graph-theory approach. Additionally, scAllele enables allele-specific splicing analysis, a unique feature not previously used at the single-cell level.
- L-GIREMI – contributor** 07/2023
Built in Python | GitHub: <https://github.com/gxiaolab/L-GIREMI>
 L_GIREMI uses a robust statistical approach based on mutual information for the detection of A-to-I RNA editing in long-read RNA sequencing data unveiling insights about the distribution of these sites on individual RNA molecules.
- BEAPR – contributor** 03/2019
Built in C++, R | GitHub: <https://github.com/gxiaolab/BEAPR>
 BEAPR identifies allele-specific binding events in eCLIP-Seq data to functionally characterize genetic variants. Taking into account crosslinking-induced sequence propensity and variations between replicated experiments, BEAPR achieves high accuracy calls and identifies clinically relevant variants.

COMMUNITY ENGAGEMENT AND SERVICE

- Diversity and Science Lectures (DASL)** 06/2024 – 08/2024
 Member of the DASL organization at UCLA that aims to promote diversity and inclusion through scientific seminars, social activities, and workshops.
- Reviewer for the United World Colleges (UWC)** 10/2024
 Reviewed applications for the Bolivian UWC selection committee, contributing to the evaluation and selection of scholarship recipients. 10/2023
 10/2022
- UCLA Latino Alumni Association** 10/2022 – 06/2023
 Guided undergraduate Latino students in strategizing their paths toward graduate school or securing employment opportunities.
- Dashew Center for International Students and Scholars** 09/2016 – 06/2017
 Assisted incoming international students to help them adapt to the university and life in the United States
- Engineering Graduate Student Association (eGSA)** 09/2015 – 12/2021
- Health-Oriented Professional Engineering Students (HOPES)** 09/2014 – 05/2015

- **Society of Hispanic Professional Engineers (SHPE)**

09/2012 – 05/2015

SKILLS

- **Computational skills**
 - Analysis of Next Generation Sequencing (NGS) data, including bulk RNA-seq, single-cell RNA-Seq, and Third Generation Sequencing data (PacBio, Nanopore)
 - Software development using Python
 - Software packaging and distribution (PyPI, Apptainer, GitHub)
 - Machine Learning in Genomics using Scikit-Learn
 - Programming languages: proficient with Python, Bash, and R
 - Carried out projects on Google Cloud Platform (GCP)
 - Operating Systems: Linux/Unix/Ubuntu, Mac OS, Windows
- **Languages**
 - Native or bilingual language proficiency: Spanish and English.
 - Limited working proficiency: French and Italian